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Design Specification for VCS-3

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Revision History

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Table of Contents

1	Introduction.....	5
2	Data Sheets.....	5
3	Acronyms and Abbreviations	6
4	Functional Description	7
4.1	Block Diagram	7
4.2	Module Description.....	8
4.3	Functional Description.....	8
4.3.1	Xilinx UltraScale+ MPSoC	8
4.3.2	LPDDR4	8
4.3.3	eMMC.....	8
4.3.4	SPI ROM.....	9
4.3.5	Camera FPC	9
4.3.6	IMU	9
4.3.7	UART.....	9
4.3.8	GPIO	9
4.3.9	CANBus	9
4.3.10	USB	10
4.3.11	Power Supply	10
4.3.12	Clocks.....	11
4.3.13	MCU Booting	11
4.3.14	LEDs	11
4.3.15	Test Points.....	12
5	Verification & Validation Procedures.....	12
6	Review Procedures	12
7	Circuit Description / Diagrams.....	13
8	Footprint.....	21
8.1	Top View.....	21
8.2	Bottom View.....	22
8.3	3D View.....	23
9	Device Pinouts	24
9.1	Zynq UltraScale+ Pinout.....	24
9.2	MCU Pinout	30
10	Connector Pinouts.....	31
10.1	Camera.....	31
10.2	CAN Bus (UART1)	31

10.3	GPIO (UART2)	32
10.4	External Power (UART0).....	32
10.5	JTAG	32
11	Physical Properties	33
12	Safety	33
13	EMC.....	33
14	Ordering Information.....	33
15	Compliance Matrix.....	34

1 Introduction

The VCS-3 is a standalone autonomy solution, providing a streamlined means for the robot integrator to design AI algorithms and IP into their autonomous systems.

2 Data Sheets

Xilinx Zynq UltraScale+ MPSoC Data Sheet:

<https://docs.xilinx.com/v/u/en-US/ds891-zynq-ultrascale-plus-overview>

eMMC datasheet

https://www.issi.com/WW/pdf/IS21_22ES08G.pdf

SPI ROM datasheet

https://media-www.micron.com/-/media/client/global/documents/products/data-sheet/nor-flash/serial-nor/mt25q/die-rev-b/mt25q_qlkt_u_512_abb_0.pdf

USB controller datasheet

<https://www.mouser.co.uk/datasheet/2/268/332x-467932.pdf>

LPDDR4 datasheet

<https://www.micron.com/products/dram/lpdr4/part-catalog/mt53d512m32d2ds-053-aut>

PLL datasheet

<https://www.renesas.com/us/en/document/dst/5p49v6901-datasheet>

IMU datasheet

https://invensense.tdk.com/wp-content/uploads/2016/06/DS-000189-ICM-20948-v1.3.pdf?ref_disty=digikey

DCDC datasheets

https://product.tdk.com/en/system/files?file=dam/doc/product/power/switching-power/micro-pol/data_sheet/fs1406_datasheet.pdf

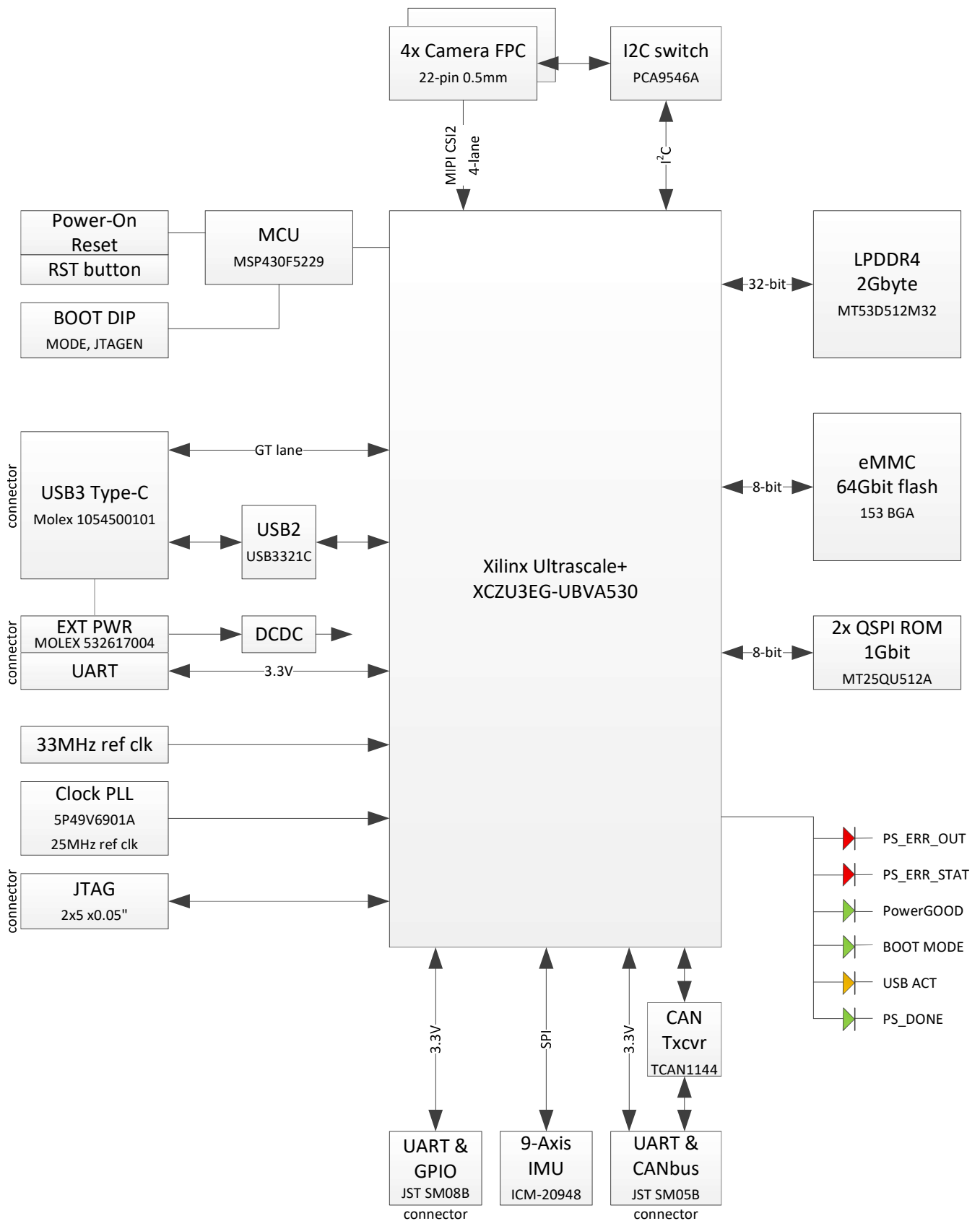
<https://www.murata.com/products/productdata/8807035797534/MYRGM-b-w.pdf?1583754814000>

3 Acronyms and Abbreviations

BER	Bit error rate.
bps	Bits per second.
CAN	Controller Area Network. Automotive serial bus.
DDR	Double Data Rate. Typically in reference to memories.
eMMC	embedded Multi-Media Card. Flash memory device.
ESD	Electrostatic Discharge.
FPC	Flexible Printed Circuit.
FPGA	Field Programmable Gate Array.
Gbps	Giga (10 ⁹) bits per second.
GPIO	General Purpose I/O.
GTX	High speed serial interface.
I ² C	Inter-Integrated Circuit. Multi-master, multi-slave, serial bus.
IMU	Inertial Measurement Unit. Accelerometer.
JTAG	Joint Test Action Group. Serial interface for chip testing.
LED	Light Emitting Diode
LPDDR4	Low Power DDR memory, 4 th generation.
MCU	Microcontroller Unit
MGT	See GTX.
MPSoC	Multi-Processor System on a Chip.
PLL	Phase Locked Loop.
PSU	Power Supply Unit.
ROM	Read Only Memory.
SoC	System on Chip.
SPI	Serial Peripheral Interface.
UART	Universal Asynchronous Receiver Transmitter.
ULPI	UTMI Low Pin Interface.
USB	Universal Serial Bus.
UTMI	USB Transceiver Macrocell Interface.
VBUS	USB power signal/pin.
VCS	Video Controls Sensors.

4 Functional Description

4.1 Block Diagram



4.2 Module Description

The VCS-3 utilizes a Xilinx UltraScale+ MPSoC coupled with high speed LPDDR4 memory to produce a highly compact evaluation platform. Together with 4 digital camera interfaces, a 9-axis IMU, and a CANBus interface, this platform is ideally suited for autonomous machine research.

Device booting can be from either SPI ROMs or eMMC flash, with no bulky, fiddly or unreliable SD cards.

Numerous on-board power rails are generated from a single external 5V supply, or via a USB3 Type-C interface.

Several LEDs indicate board functionality, and numerous test points allow access to the various interfaces.

4.3 Functional Description

4.3.1 Xilinx UltraScale+ MPSoC

This is an XCZU3EG device in a UBVA530 BGA package. It requires 13 power rails (some of them common). Refer to the power supply section, page 10.

Device booting is controlled by the microcontroller, page 11, and can be from eMMC or SPI ROM.

A 33MHz clock is used as the system clock, and a separate PLL based clock used for the GTX. See page 11.

The UltraScale+ device interfaces are detailed following.

The pin-out is given in the pinout section, page 24.

4.3.2 LPDDR4

A single 32-bit wide MT53D512M32D2DS device provides 2Gbytes of memory.

This device connects directly to the PS side of the UltraScale+. Termination is provided on-die with the exception CKE (clock enable) which use discrete resistors at the track ends.

The LPDDR4 is powered at 1.8V and 1.1V internal, with 1.1V I/O.

4.3.3 eMMC

This flash memory provides 8Gbytes of non-volatile storage in a 153 BGA package.

It connects directly to the MIO pins of the PS.

It is powered at 3.3V with 1.8V I/O.

4.3.4 SPI ROM

Two 4-bit wide MT25QU512 devices form an 8-bit SPI ROM interface to the UltraScale+ on its PS interface.

Each device is 512Mbits thus the two provide 1Gbit of storage.

They are powered at 1.8V.

4.3.5 Camera FPC

Four CSI-2 4-lane interfaces are provided over separate 22 pin 0.5mm FPC. The interfaces' differential signals are routed as high-speed nets (length and skew matched) to the UltraScale+ on its PL bus. Each lane supports data rates up to 900Mb/s.

The pin-out is shown here: <https://www.arducam.com/raspberry-pi-camera-pinout/>

The camera interface I2C connection is supplied via a PCA9546A-RGY I2C switch. This has the ability to access cameras individually or via a broadcast to all cameras simultaneously. The I2C is connected to the PS interface.

The camera clock is from the PL interface via a potential divider to ensure the voltage is less than 2.8V.

The cameras are powered at 3.3V.

4.3.6 IMU

An ICM-20948 9-axis MEMS device is connected as an SPI device to the PS interface.

It is powered at 3.3V and the interface includes an interrupt.

Alternative parts, e.g. MPU9250, may be fitted as they have compatible footprints.

4.3.7 UART

Three UART interfaces are provided; one on the CANbus connector (JST SM05B-SRSS-TB(LF)(SN)), one on the GPIO connector (JST SM08B-SRSS-TB(LF)(SN)), and the third on the external power input connector (MOLEX 532617004).

UARTs 0 & 1 are provided by MIO pins on the PS interface. AUX_UART2 is via the PL interface. All 3 are at 3.3V levels.

4.3.8 GPIO

Four GPIO signal are connected to the PL interface at 3.3V, and share a connector with the AUX_UART2.

4.3.9 CANBus

A TCAN1144DMTRQ1 CAN transceiver supports data rates up to 5Mb/s. This is connected to the PS as an SPI device.

This device has both 5V and 3.3V power supplies.

4.3.10 USB

USB3 signalling is provided directly by the UltraScale+ device using a single PS MGT lane.

The MGT clock is provided by a PLL based clock generator, page 11.

USB2 signalling is via a USB3321C controller. This part connects to the UltraScale+ as an 8-bit ULPI interface. The ULPI reference clock, 60MHz, is also provided by the PLL.

The Type-C connector's USB3 signals are switched using a CBTU02043HEJ USB switching device. This device selects the connector's active differential pairs. [USB Type-C allows for cable insertion in either orientation, thus two sets of Rx/Tx pairs are available.]

ESD protection is provided for all of the Type-C's signals.

Power to the VCS-3 module may be derived from the USB connector, or power from the external power connector can be provided to the USB connector. Whether this module is a USB host or device is determined by the MCU which in turn controls the required power switches.

4.3.11 Power Supply

Several TDK FS1406-0600 DCDC converters are used to generate a range of local power rails. A single MYRGM330150 DCDC generates the local 3.3V rail. The DCDCs are supplemented by a couple of linear regulators (TPS72012) and further power switches (SIP32455) to ensure the power rail sequencing is correct for the UltraScale+.

Rail	Voltage	Device	Rail	Voltage	Device
PL_VCCINT	0.72V	FS1406	PS_VCCINT, PS_LP0V85	0.85V	FS1406
1V8	1.8V	FS1406	PS_FP0V85	0.85V	FS1406
PL_1V2	1.2V	TPS72012	VLPDDR4	1.1V	FS1406
PS_PLL	1.2V	TPS72012	VCCO3V3	3.3V	MYRGM330150
PS_AVCC	0.85V	FS1406	PS_AVTT	1.8V	FS1406
VCCINT_IO	0.85	SIP32455	VCCAUX	1.8V	SIP32455

Both the PS_VCCINT and PL_VCCINT DCDC converters are connected to the PS via I2C. This allows for the UltraScale+ to adjust these voltage rails (minimum 0.6V).

The PL_VCCINT power rail can be disabled using MIO pin MIO42 (pin J9).

The PS_AVCC power rail can be disabled using MCU pin A6 (port P5.5).

4.3.12 Clocks

A 33.33MHz oscillator provides the UltraScale+ reference clock (PS_REF_CLK_503, pin B25).

A Renesas 5P49V6901A000NLGI PLL can be configured using the I2C bus in order to provide the USB GTX clock (PLL OUT1). The PLL reference clock is 25MHz.

The PLL also produces the 60MHz USB2 reference clock on OUT4.

4.3.13 MCU Booting

A Texas Instruments MSP430F5229IYFFT MCU is programmed to control the UltraScale+ boot mode, reset and USB3 port direction.

Boot mode is selected by 2 dip switches. The MCU uses the state of these switches to control the UltraScale+ bootmode using the BMODE pins.

BMODE Pins [3:0]	Function
0000	PS JTAG
0001	QSPI (24 bit)
0010	QSPI (32 bit)
0110	eMMC

The MCU is programmed using a 2-wire interface accessible on surface mount pads compatible with spring-loaded pins.

The MCU also monitors the power good signals from the DDR and MGT regulators, and in turn illuminates a power good LED.

Overall system reset is via a TPS3899DL01DSER voltage supervisor. This has a reset input from a push button switch.

Refer to full circuit diagrams for full connectivity.

The device pinout is shown on page 30.

4.3.14 LEDs

Six LEDs are grouped together and indicate the following:

LED	Colour	Function
LED5	Green	PS_DONE
LED4	Red	PS_ERROR_OUT
LED6	Red	PS_ERROR_STATUS
LED3	Orange	USB activity
LED1	Green	Power good
LED2	Green	Boot mode

4.3.15 Test Points

Test Point	Function
1	PORT (USB)
2	1.8V
3	SBWTCK
4	SBWTDIO
5	GND
6	BMODE0
7	BMODE1

Test Point	Function
8	BMODE2
9	BMODE3
10	CAM_SCL
11	CAM_SDA
12	ZYNQ-RST#
13	SDA (MCU)
14	SCL (MCU)

5 Verification & Validation Procedures

In accordance with Sundance Quality Procedures.

6 Review Procedures

In accordance with Sundance Quality Procedures.

10 Connector Pinouts

10.1 Camera

Pin	Function
1	GND
2	D0_N
3	D0_P
4	GND
5	D1_N
6	D1_P
7	GND
8	CK_N
9	CK_P
10	GND
11	D2_N
12	D2_P
13	GND
14	D3_N
15	D3_P
16	GND
17	EN
18	CLK (2.8V)
19	GND
20	SCL
21	SDA
22	3.3V

10.2 CAN Bus (UART1)

Pin	Function
1	RX
2	TX
3	GND
4	CAN H
5	CAN L

10.3 GPIO (UART2)

Pin	Function
1	3.3V
2	RX
3	TX
4	GPIO0
5	GPIO1
6	GPIO2
7	GPIO3
8	GND

10.4 External Power (UART0)

Pin	Function
1	EXT 5V IN
2	RX
3	TX
4	GND

10.5 JTAG

Function	Pin	Pin	Function
VTref	1	2	TMS
GND	3	4	TCK
GND	5	6	TDO
NC	7	8	TDI
NC	9	10	/SRST

11 Physical Properties

Dimensions	50mm x 30mm
Weight	TBA.
Supply Voltage	+5V $\pm$ 5%
Supply Current	3A max.
MTBF	

12 Safety

This module presents no hazard to the user when in normal use, unless it's in a drone and flies into your face!

13 EMC

This module is designed to operate from within an enclosed host system, which is built to provide EMC shielding. Operation within the UK/EU EMC guidelines is not guaranteed unless it is installed within an adequate host system.

This module is protected from damage by fast voltage transients originating from outside the host system which may be introduced through the output cables.

Short circuiting any output to ground does not cause a host PC system to lock up or reboot.

14 Ordering Information

Part number: VCS-3

15 Compliance Matrix

Feature	Spec Reference	Implementation
UltraScale+	4.3.1	U2, XCZU3EG-UBVA530.
LPDDR4	4.3.2	U1, MT53D512M32D.
eMMC	4.3.3	U3.
SPI ROM	4.3.4	U4 & U8, MT25QU512A.
Camera FPC	4.3.5	J1, 3, 4 & 5. I2C U6, PCA9546A.
IMU	4.3.6	U5, ICM-20948.
UART	4.3.7	J7, J8 & J9.
GPIO	4.3.8	J8.
CAN Bus	4.3.9	U9, TCAN1144D. J7.
USB	4.3.10	U2 (GTX), U20, USB3321C.
Power supply	4.3.11	U7, 13, 17, 22, 34, 15, 10, 23, 14, 16, 24 & 25.
Clocks	4.3.12	X1, U11, 5P49V6901, X2.
MCU / Booting	4.3.13	SW2, U26, SW1, U21, MSP430F5229.
LEDs	4.3.14	LED1-6.
Test Points	4.3.15	As described.